

Online Appendix

The Market Pricing of Proprietary AI Model Capability

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1 Sample Construction

The starting point is a set of 60 major AI model release events between January 2024 and March 2026. The event screen requires an official model or model-family release, a confirmed public release date, public access or benchmark visibility, and an economic link to at least one AI-related listed firm. The clean firm-event panel has 5,160 observations and 86 firms. The main capability sample has 47 releases with nonmissing Artificial Analysis (AA) Intelligence Index scores and 3,780 complete firm-event observations. The closed-source sample has 36 events and 2,899 observations. The open-weight sample has 11 events and 881 observations.

2 Variable Definitions

The dependent variable is a cumulative abnormal return. The market-model CAR uses a firm-level market model fitted before the release date. The FF3 CAR uses the Fama-French three-factor model. The main window is $[0, +20]$ trading days. The pre-event CAR uses the pre-release window available in the cleaned panel. The main capability measure is the AA

Table A1: Sample summary

Sample component	Observations	Events
Clean firm-event panel	5,160	60
Main capability sample	3,780	47
Closed-source capability sample	2,899	36
Open-weight capability sample	881	11
Pre-event CAR sample	3,762	47
FF3 CAR sample	3,782	47
US-traded ticker screen	2,857	47

The main capability sample requires nonmissing AA Intelligence Index, CAR[0, +20], firm controls, year, event identifier, and firm identifier. The US-traded ticker screen uses a rule-based ticker filter and should be read as a robustness check rather than a formal exchange-country classification.

Intelligence Index. The open-weight indicator equals one when the model is released under an open-weight or open-source regime.

Firm-event relationship variables are binary indicators. They include owner, investor, cloud, business upstream, real upstream, business downstream, real downstream, and competitor. The relationship variables were merged into the cleaned panel using event and firm identifiers. They are used for heterogeneity and appendix checks, not for the baseline result.

3 Main Results with Small-Cluster Inference

Table A2 reports the core estimates with CR0, CR2, and wild cluster bootstrap inference. The closed-source CAR[0, +20] result is the most stable estimate. The full-sample slope and open-weight interaction are weaker under the most conservative inference and should be read as supporting evidence.

Table A2: Core estimates with event-cluster inference

Specification	N	Events	Coef.	SE	CR2 p	Wild p
Full sample, CAR[0, +20]	3,780	47	0.00152	0.00067	0.054	0.053
Closed-source, CAR[0, +20]	2,899	36	0.00232	0.00062	0.007	0.008
Interaction, closed-source slope	3,780	47	0.00226	0.00060	0.005	0.008
Interaction, open-weight term	3,780	47	-0.00373	0.00120	0.048	0.086
Investor interaction	3,780	47	-0.00272	0.00098	0.050	0.126
Cloud interaction	3,780	47	-0.00213	0.00104	0.100	0.137
Owner interaction, CAR[0, +1]	3,763	47	-0.00178	0.00072	0.059	0.034
Owner interaction, CAR[0, +2]	3,768	47	-0.00177	0.00070	0.052	0.046

The coefficient is on the AA Intelligence Index or the stated interaction. Wild p -values use 4,999 Rademacher draws clustered by event. The investor, cloud, and owner rows are exploratory relationship checks.

4 Firm Fixed Effects, Alternative Returns, and Pre-Event CARs

Table A3 reports robustness checks that address firm composition, alternative abnormal-return models, pre-event pricing, and the exchange-country mix in the firm panel. The closed-source coefficient remains positive in all post-event specifications. The pre-event coefficient is not statistically significant.

Table A3: Robustness checks for the capability slope

Specification	N	Events	Coef.	SE	p
All, firm fixed effects, event cluster	3,780	47	0.00154	0.00067	0.026
All, firm fixed effects, two-way cluster	3,780	47	0.00154	0.00077	0.052
Closed-source, firm fixed effects, event cluster	2,899	36	0.00226	0.00063	0.001
Closed-source, firm fixed effects, two-way cluster	2,899	36	0.00226	0.00087	0.014
All, FF3 CAR[0, +20]	3,782	47	0.00102	0.00044	0.025
Closed-source, FF3 CAR[0, +20]	2,901	36	0.00118	0.00051	0.027
All, pre-event CAR	3,762	47	0.00016	0.00028	0.576
Closed-source, pre-event CAR	2,884	36	0.00047	0.00030	0.119
All, US-traded ticker screen	2,857	47	0.00193	0.00080	0.019
Closed-source, US-traded ticker screen	2,191	36	0.00271	0.00072	0.001

The table reports the coefficient on the AA Intelligence Index. Two-way clustering clusters by event and firm. The US-traded ticker screen is based on ticker format and is used only as a robustness check.

5 Event-Level Aggregation

The main panel repeats the event-level capability score across firms. Table A4 collapses the data to the event level by averaging CARs within each release. The closed-source result survives when all firms are used in the equal-weighted event mean. The result is weaker when only firms with explicit relationship indicators are averaged.

Table A4: Event-level aggregation

Event-level outcome	Events	Coef.	SE	<i>p</i>
All firms, all releases	47	0.00099	0.00062	0.120
All firms, closed-source releases	36	0.00172	0.00067	0.016
Relationship firms, all releases	47	0.00061	0.00078	0.437
Relationship firms, closed-source releases	36	0.00142	0.00103	0.178

Each row estimates an event-level regression in which the dependent variable is an equal-weighted mean CAR[0, +20] within the release event. Relationship firms are firm-event pairs with at least one explicit relationship indicator.

6 Leave-One-Creator-Out

Table A5 excludes major model creators one at a time. The closed-source coefficient remains positive and statistically significant in each reported exclusion.

Table A5: Leave-one-creator-out, closed-source releases

Excluded creator	N	Events	Coef.	SE	<i>p</i>
Google	1,927	24	0.00204	0.00077	0.015
OpenAI	2,010	25	0.00426	0.00069	0.000
Anthropic	2,272	28	0.00198	0.00062	0.004
Alibaba	2,820	35	0.00215	0.00061	0.001
Meta	2,816	35	0.00235	0.00061	0.001
xAI	2,733	34	0.00225	0.00066	0.002
Microsoft	2,816	35	0.00207	0.00070	0.006

The table reports the closed-source coefficient on the AA Intelligence Index after excluding all events from the listed creator.

7 Specification-Curve Evidence

Table A6 compares alternative capability measures across reasonable specifications. The AA Intelligence Index has the strongest overall pattern among the LLM capability measures. The AA Media Elo measure behaves differently and is concentrated in short-window responses.

Table A6: Specification-curve summary by capability measure

Capability measure	Specifications	5% significant	Positive	Median coefficient
AA Intelligence Index	252	23.8%	66.3%	0.00020
AA Media Elo	231	20.3%	62.3%	0.00002
AA Coding Index	252	11.5%	51.6%	0.00001
AA Math Index	231	5.6%	26.8%	-0.00009

The specification curve varies the event window, control set, subsample, and year fixed effects. The table reports the share of estimates significant at 5 percent using the corresponding clustered standard error in each specification.

Table A7: AA Intelligence Index across event windows

Window	5% significant	Median coefficient
CAR[0, +1]	30.6%	0.000124
CAR[0, +2]	25.0%	0.000091
CAR[0, +3]	22.2%	0.000080
CAR[0, +5]	2.8%	-0.000014
CAR[0, +10]	13.9%	-0.000247
CAR[0, +15]	25.0%	0.000507
CAR[0, +20]	47.2%	0.000848

8 Additional Heterogeneity

8.1 Open-Weight and Closed-Source Specification Patterns

The long-window open-weight estimates are negative in point estimates, while the closed-source specifications are positive. Because the open-weight group has only 11 events, the main text treats the interaction result as suggestive mechanism evidence.

Table A8: Open-weight and closed-source patterns

Subsample	5% significant	Positive	Median coefficient
Open-weight releases	38.1%	7.1%	-0.000599
Closed-source releases	26.2%	88.1%	0.000512

8.2 Owner Firms

Owner firms display a short-window negative response that dissipates by 10 to 20 trading days. This is an exploratory result because the owner sample is small.

Table A9: Owner-firm short-window response

Window	Coef.	SE	<i>p</i>
CAR[0, +1]	-0.00232	0.00068	0.003
CAR[0, +2]	-0.00287	0.00063	0.000
CAR[0, +3]	-0.00270	0.00077	0.002
CAR[0, +5]	-0.00205	0.00102	0.057
CAR[0, +10]	-0.00080	0.00201	0.695
CAR[0, +20]	0.00047	0.00285	0.871

8.3 Event Tier

Tier 2 events show a larger capability slope than Tier 1 events. The interaction between intelligence and Tier 1 is not statistically significant, so we treat this as a descriptive split.

Table A10: Capability slope by event tier

Tier	N	Events	CAR[0, +20] coefficient	<i>p</i>
Tier 1	1,438	18	0.00153	0.045
Tier 2	2,159	27	0.00374	0.000

8.4 Calendar Year

The capability slope is not significant in 2024 and turns positive in 2025. The 2026 estimate is large but imprecise because the sample is small.

Table A11: Capability slope by release year

Year	N	Coef.	SE	<i>p</i>
2024	1,291	-0.00044	0.00060	0.474
2025	2,183	0.00206	0.00087	0.026
2026	306	0.01860	0.01157	0.206

8.5 Industry

The strongest industry-level result appears in internet services and infrastructure. Software and semiconductors show positive but more marginal patterns.

Table A12: Industry-level capability slopes

Industry	N	Coef.	CR0 <i>p</i>	CR2 <i>p</i>
Internet services and infrastructure	235	0.00190	0.005	0.015
Software	1,330	0.00295	0.045	0.078
Semiconductors and semiconductor equipment	606	0.00212	0.032	0.060
Technology hardware, storage, and peripherals	348	-0.00083	0.177	0.210
IT services	325	0.00047	0.533	0.561
Internet retail	184	-0.00004	0.947	0.950
Other or non-IT	752	0.00027	0.567	0.600

8.6 Mag7 and Non-Mag7 Firms

The Mag7 and non-Mag7 slopes are similar. The interaction between intelligence and the Mag7 indicator is not statistically significant.

Table A13: Mag7 and non-Mag7 slopes

Group	N	Coef.	CR0 <i>p</i>	CR2 <i>p</i>
Mag7	328	0.00170	0.045	0.081
Non-Mag7	3,452	0.00147	0.037	0.068

9 Media Sentiment

FinBERT sentiment is negatively related to $CAR[0, +20]$. In joint models, the sentiment and capability coefficients retain their signs but are marginal under CR2. The sentiment-capability interaction is not statistically significant.

Table A14: Media sentiment and $CAR[0, +20]$

Specification	Coef.	CR0 <i>p</i>	CR2 <i>p</i>
FinBERT sentiment, ± 5 days	-0.081	0.000	0.003
FinBERT sentiment, ± 20 days	-0.096	0.003	0.015
Sentiment in joint model	-0.061	0.024	0.057
Intelligence in joint model	0.00158	0.035	0.087

10 Event-Window Overlap

AI model releases are clustered in calendar time. Among the 47 intelligence events, only four have no other intelligence event within 20 calendar days. The median event has three other events within that window. Because a strict non-overlap sample would be too small, the main paper keeps the full capability sample and reports this diagnostic.

Table A15: Overlap within 20 calendar days

Statistic	Other intelligence events within 20 days
Minimum	0
25th percentile	2
Median	3
Mean	2.68
75th percentile	4
Maximum	6
Events with no overlap	4 of 47

11 Null and Exploratory Results

Several results are informative but should not be interpreted as main findings. The competitor subsample has a small and statistically insignificant capability slope. Real downstream firms have a larger point estimate but too few observations for a precise estimate. Investor and cloud interactions are negative but do not survive the most conservative inference.

Table A16: Exploratory and null relationship results

Result	N	Events	Coef.	<i>p</i>
Competitor subsample	366	47	0.00052	0.352
Real downstream subsample	62	20	0.00200	0.235
Investor interaction, wild bootstrap	3,780	47	-0.00272	0.126
Cloud interaction, wild bootstrap	3,780	47	-0.00213	0.137